WORKED SOLUTION

STEP II 2010, Question 9

Mechanics • Colliding Projectiles

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Read the question before you go on. It is in the STEP Support Programme's question database:

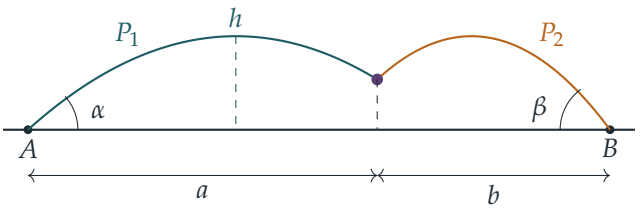
step.maths.org/questions/10-s2-q9

The key idea

Two particles projected at the same moment fall with the same acceleration g, so the only way their heights can agree at the moment of collision is for their initial vertical velocities to be equal. Then their vertical motions are identical: same greatest height h, reached at the same time, and both would land at the same time. The collision happens between the top of the flight and the landing, and the horizontal distance each particle has covered is proportional to the time, which pins it between two simple bounds.

Solution

We model both particles as projectiles moving freely under gravity g, with no air resistance. Let P_1 have speed u and P_2 have speed v, take $t = 0$ at projection and let the collision happen at time $t_c > 0$.



Equal vertical motions

At time t the heights of P_1 and P_2 are

$$y_1 = ut\sin\alpha - \tfrac{1}{2}gt^2, \qquad y_2 = vt\sin\beta - \tfrac{1}{2}gt^2.$$

At the collision $y_1 = y_2$, so $(u\sin\alpha - v\sin\beta)\,t_c = 0$, and since $t_c > 0$,

$$u\sin\alpha = v\sin\beta.$$

So the two particles have exactly the same vertical motion. In particular both reach the same greatest height h at the same time T, where

$$T = \frac{u\sin\alpha}{g} = \frac{v\sin\beta}{g}, \qquad h = \frac{u^2\sin^2\alpha}{2g} = \frac{v^2\sin^2\beta}{2g},$$

and both would return to the ground at time $2T$.

Ask yourself What do "in the air" and "after P_1 has attained its maximum height" tell us about t_c?

P_1 is at its highest at time T, so "after" gives $t_c > T$. Both particles are above the ground only for $0 < t < 2T$, so a collision in the air needs $t_c < 2T$. Hence

$$T < t_c < 2T.$$

The bound on b

Horizontally, P_2 moves with constant speed $v\cos\beta$, so $b = v\cos\beta\, t_c$. At time T it would have covered

$$v\cos\beta\, T = \frac{v^2\sin\beta\cos\beta}{g} = \frac{v^2\sin^2\beta}{2g}\cdot\frac{2\cos\beta}{\sin\beta} = 2h\cot\beta.$$

Therefore

$$b = v\cos\beta\, T\cdot\frac{t_c}{T} = 2h\cot\beta\cdot\frac{t_c}{T},$$

and since $1 < t_c/T < 2$ and $2h\cot\beta > 0$ (as β is acute),

$$2h\cot\beta < b < 4h\cot\beta,$$

as required.

The bound on a

In exactly the same way, $a = u\cos\alpha\, t_c$ and

$$u\cos\alpha\, T = \frac{u^2\sin\alpha\cos\alpha}{g} = 2h\cot\alpha,$$

so $a = 2h\cot\alpha\cdot t_c/T$, and $1 < t_c/T < 2$ gives

$$2h\cot\alpha < a < 4h\cot\alpha.$$

This uses the same h: h is the greatest height of P_1 by definition, and it is also the greatest height of P_2 because the vertical motions are identical.

Checks

- Boundary cases: a collision exactly at the top ($t_c = T$) gives $a = 2h\cot\alpha$, the familiar half-range $\frac{u^2\sin 2\alpha}{2g}$; a collision at the moment of landing ($t_c = 2T$) gives $a = 4h\cot\alpha$, the full range $\frac{u^2\sin 2\alpha}{g}$. The question's strict inequalities sit between these.

- Numerically: with $g = 9.8$, $u = 20$, $\alpha = 0.7$, $\beta = 0.9$ (radians), $v = u \sin\alpha / \sin\beta$ and $t_c = 1.5T$, we get $h \approx 8.470$ and $a \approx 30.17$, which lies between $2h\cot\alpha \approx 20.11$ and $4h\cot\alpha \approx 40.22$ (indeed it is exactly 1.5×20.11). We tested 20 000 random set-ups this way, stepping P_1's motion to confirm h and checking that the particles really coincide.

Common slips

- Treating the times to reach the collision point as different for the two particles: they are projected simultaneously, so it is the same t_c.
- Forgetting to justify the upper bound: it comes from the collision being in the air, that is, before the common landing time $2T$.

Beyond STEP

The key step is a change of frame: in a frame falling freely with acceleration g, both particles move in straight lines at constant velocity, so they collide only if they were aimed at each other in that frame, which is the classic "monkey and hunter" idea. Thinking in accelerating frames, and the fictitious forces they bring when the acceleration is not uniform, is a central theme of first-year mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats projectiles, relative motion and non-inertial frames with many good problems.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.